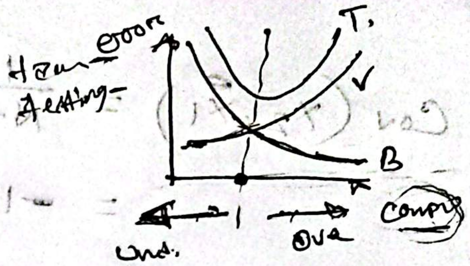
Step-1

(A) $\bar{x}_1 = \dfrac{4+8+13+7}{4} = 8$

$\bar{x}_2 = \dfrac{11+4+5+14}{4} = 8.5$



(B)

Feature Index	x_1	x_2	$x_1-\bar{x}_1$ (8)	$x_2-\bar{x}_2$ (8.5)	$(x_1-\bar{x}_1)(x_2-\bar{x}_2)$
1	4	11	-4	2.5	-10
2	8	4	0	-4.5	0
3	13	5	5	-3.5	-17.5
4	7	14	-1	5.5	-5.5
N=4				0	-33

Step: 2 Covariance matrix, $S = \begin{bmatrix} Cov(x_1, x_1) & Cov(x_1, x_2) \\ Cov(x_2, x_1) & Cov(x_2, x_2) \end{bmatrix}$

$$Cov(x_1, x_1) = \frac{1}{N-1}\sum (x_1-\bar{x}_1)(x_1-\bar{x}_1)$$

$$= \frac{1}{N-1}\sum (x_1-\bar{x}_1)^2$$

$$= \frac{1}{4-1}\left\{(-4)^2+0^2+5^2+(-1)^2\right\} = \frac{1}{3}\times 42 = 14.$$

$$Cov(x_1, x_2) = \frac{1}{3}\sum (x_1-\bar{x}_1)(x_2-\bar{x}_2)$$

$$= \frac{1}{3}(-10+0-17.5-5.5) = \frac{1}{3}\times(-33)$$

$$= -11.$$

$$Cov(x_2, x_1) = \frac{1}{N-1} \sum (x_2 - \bar{x}_2)(x_1 - \bar{x}_1)$$
$$= -11.$$

$$Cov(x_2, x_2) = \frac{1}{N-1} \sum (x_2 - \bar{x}_2)(x_2 - \bar{x}_2)$$
$$= \frac{1}{3} \sum (x_2 - \bar{x}_2)^2$$
$$= \frac{1}{3} \left\{ (2.5)^2 + (-4.5)^2 + (-3.5)^2 + (5.5)^2 \right\}$$
$$= \frac{1}{3} \times 69 = 23.$$

$\therefore$ Covariance matrix, $S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$.

Step:03 Estimate Eigenvalues of S.

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}.$$

The equation of S is:

$$\det(S - \lambda I) = 0 \quad \text{--- (i)}$$

I = Identity matrix $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

λ = eigenvalue

S = cov. matrix.

det = determinant.

Now,

$$S - \lambda I = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$
$$= \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix}$$

from (i) $\longrightarrow$

$$\det(S - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (14-\lambda)(23-\lambda) - \{-11 \times (-11)\} = 0$$

$$\Rightarrow 322 - 14\lambda - 23\lambda + \lambda^2 - 121 = 0$$

$$\Rightarrow \lambda^2 - 37\lambda + 201 = 0$$

$$\therefore \lambda = \frac{37 \pm \sqrt{(-37)^2 - 4 \cdot 1 \cdot 201}}{2 \cdot 1} = \frac{37 \pm \sqrt{565}}{2}$$

$$\therefore \lambda_1 = 30.3849, \quad \lambda_2 = 6.6151.$$

Step-4 : Estimate Eigen Vectors :

equation : $(S - \lambda I)U = 0$ | U = unit vector $= \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$.

For λ_1 :

$S - \lambda_1 I = [$

$$\Rightarrow \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} (14-\lambda)u_1 - 11u_2 \\ -11u_1 + (23-\lambda)u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

1st eqn.

2nd eqn.

Use 1st equation:

$$(14-\lambda)u_1 - 11u_2 = 0$$

$$\Rightarrow (14-\lambda)u_1 = 11u_2$$

$$\Rightarrow \frac{u_1}{u_2} = \frac{11}{14-\lambda}.$$

$$\therefore u_1 : u_2 = 11 : (14-\lambda).$$

~~So, eigenvector~~: $U = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}.$

for λ_1, $U_1 = \begin{bmatrix} 11 \\ 14-\lambda_1 \end{bmatrix} = \begin{bmatrix} 11 \\ 14-30.3849 \end{bmatrix}$

$$= \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix}.$$

Normalize Unit vector:

$$\text{length} = ||U_1|| = \sqrt{x^2+y^2}.$$

$$= \sqrt{(11)^2 + (-16.3849)^2}$$

$$= 19.7348.$$

$$-11u_1 + (23-\lambda)u_2 =$$

$$\Rightarrow (23-\lambda)u_2 = 11$$

$$\Rightarrow \frac{u_2}{u_1} = \frac{11}{23-}$$

$$U = \begin{bmatrix} 23-\lambda \\ 11 \end{bmatrix}$$

$$\lambda_2,\ U_2 = \begin{bmatrix} 23-\lambda_2 \\ 11 \end{bmatrix}$$

$$e_2 = \frac{U_2}{||u_2||}$$

$$||u_2|| = \sqrt{x^2}$$

~~Now make it~~

Now, $e_1 = \dfrac{U_1}{\|U_1\|}$, apply this

$$\text{eigenvector}, e_1 = \begin{bmatrix} 11/19.7348 \\ -16.3849/19.7348 \end{bmatrix} = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}.$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}.$$

(2nd equation solve করে পাওয়া যায়)

Step-5:

Transpose means row vector for e1:

$$e_1^T = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix}$$

First Observation:

$$x_{11} = 4, \quad x_{21} = 11,$$

$$\bar{x}_1 = 8, \quad \bar{x}_2 = 8.5.$$

$$PCA = e_1^T (x - \bar{x})$$

$$\therefore \text{Centered vector} = \begin{bmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix} = \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$$

Multiply vector:

$$\begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$$

Multiply term by term:

$$= 0.5574 \times (-4) + (-0.8303)(2.5)$$

$$= -4.30535 \quad (\text{Final result})$$